

SMART INDIA HACKATHON 2026

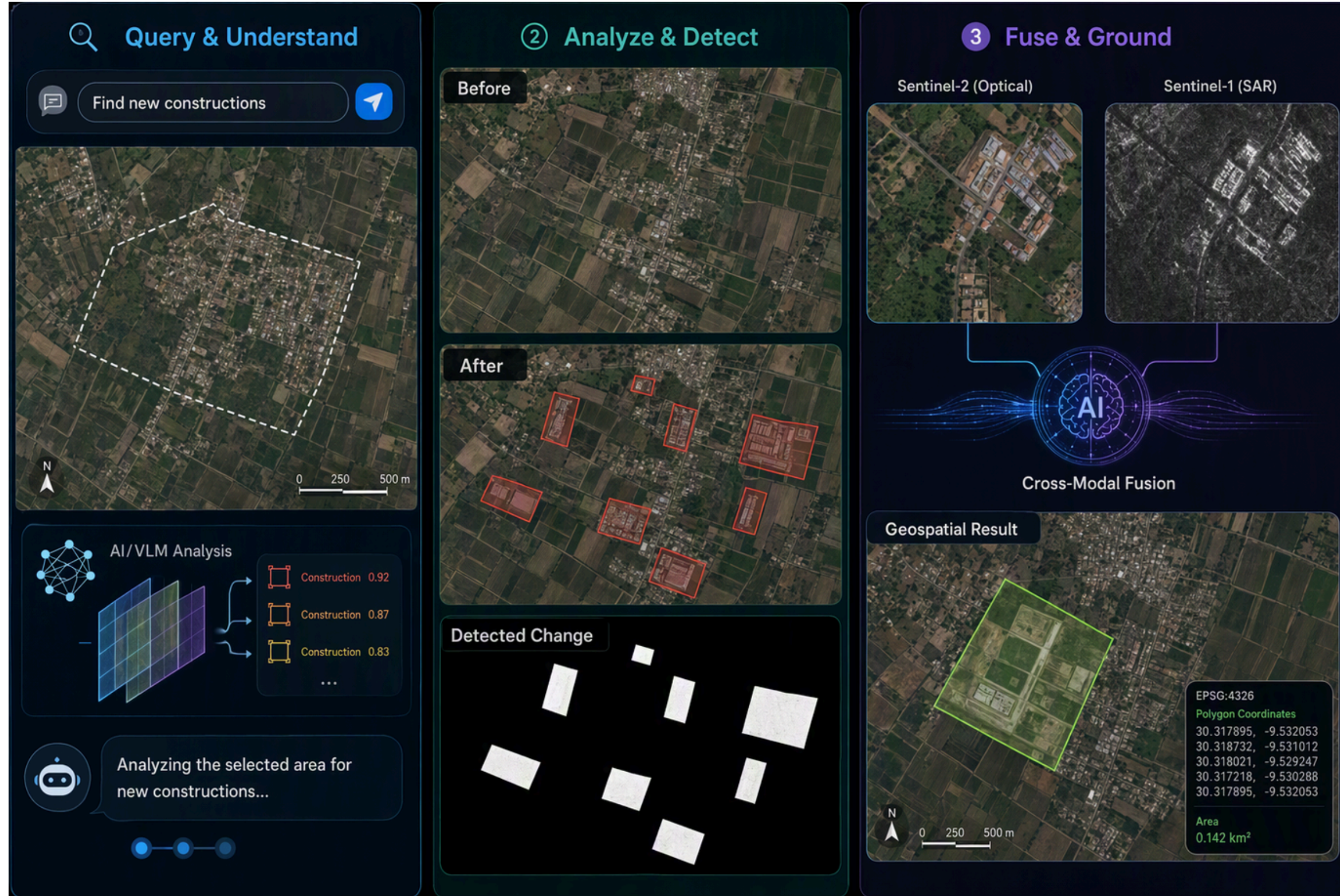


SMART INDIA
HACKATHON
2026

- **Problem Statement ID** – SIH26167
- **Problem Statement Title-** *SatQuery AI* - An Interactive Vision-Language Assistant for Multimodal Remote Sensing Image Analysis through Text Queries
- **Theme-** Space Technology
- **PS Category-** Software
- **Team ID-** 174259
- **Team Name-** Spotlight



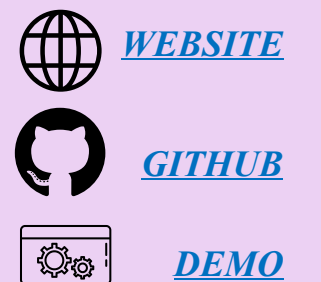
PROPOSED SOLUTION



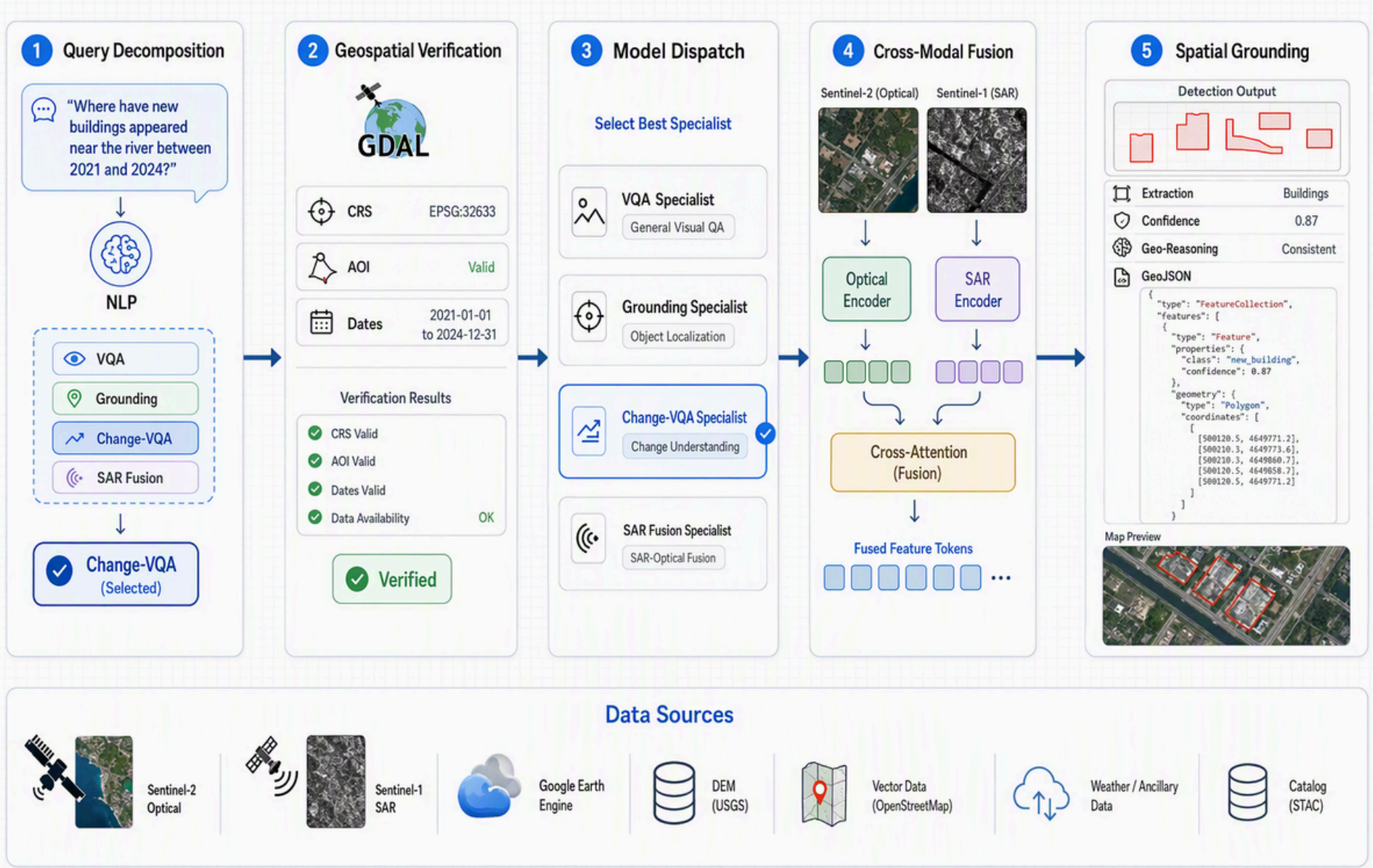
Our system lets users ask questions about satellite images in plain language, without needing advanced GIS knowledge. Instead of using one generic AI model for everything, it understands the query and automatically selects the right specialist model for tasks such as VQA, object grounding, change detection, or Optical-SAR analysis. It also checks the geospatial accuracy of the data, combines Sentinel-2 optical and Sentinel-1 SAR imagery when needed, and returns results with highlighted locations, coordinates, confidence scores, and supporting evidence. Every step is recorded, making the analysis transparent and easy to verify.

SatQuery AI uniquely combines **Optical-SAR fusion**, **bi-temporal Change VQA**, and **task-specific specialist models** within a query-driven agentic framework. Instead of relying on one generic VLM, it dynamically selects the right model for each task, enabling modular, multimodal, spatially grounded, and extensible satellite-image analysis.

SatQuery AI combines query-to-evidence intelligence, cross-sensor reasoning, and temporal change analysis to deliver spatially grounded results with visual evidence, confidence, and provenance. Its modular specialist registry enables new models and sensors to be added without redesigning the system.

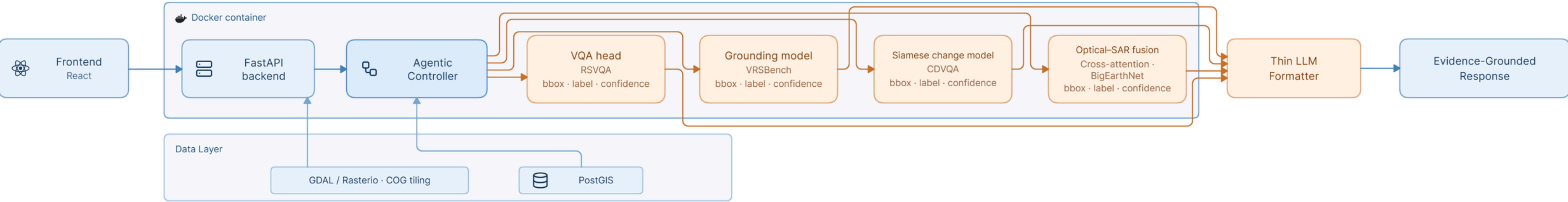


TECHNICAL APPROACH



System Workflow & Tech Stack

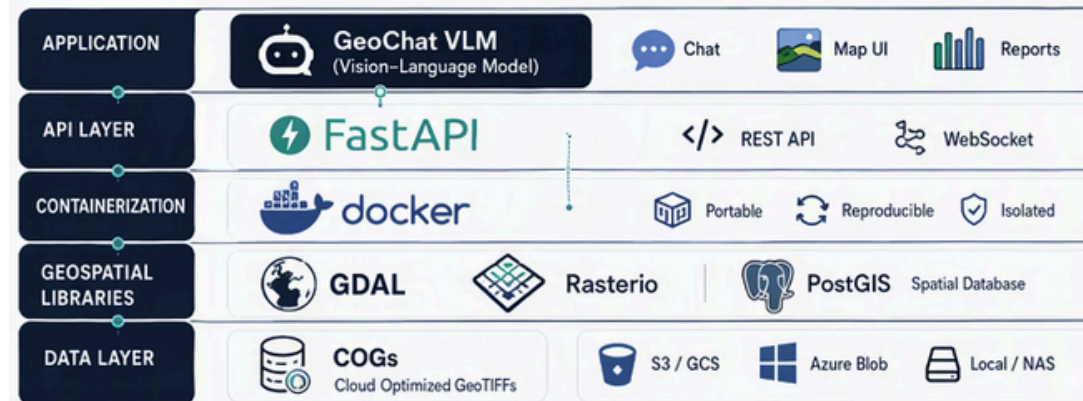
- Specialist Model Registry (per task, not one VLM):
- **VQA**: lightweight head trained on RSVQA
 - **Captioning / Grounding**: phrase-grounding model trained on VRSBench
 - **Change / Change-VQA**: Siamese change-detection network trained on CDVQA
 - **Optical-SAR Fusion**: cross-attention dual encoder trained on BigEarthNet
- Thin LLM layer**: formats each specialist's structured output (bbox, label, confidence) into a sentence, the LLM never does the visual reasoning itself
- Execution Trace**: {task, model_id, bbox, confidence, latency_ms}



Feasibility Assessment

1 OPEN & BUILDABLE STACK

Open Source • Deployable • Modular



Open standards • Open APIs • Open formats • Extensible plug-ins

2 FREE EARTH OBSERVATION DATA

No License Cost • Widely Available • Global Coverage



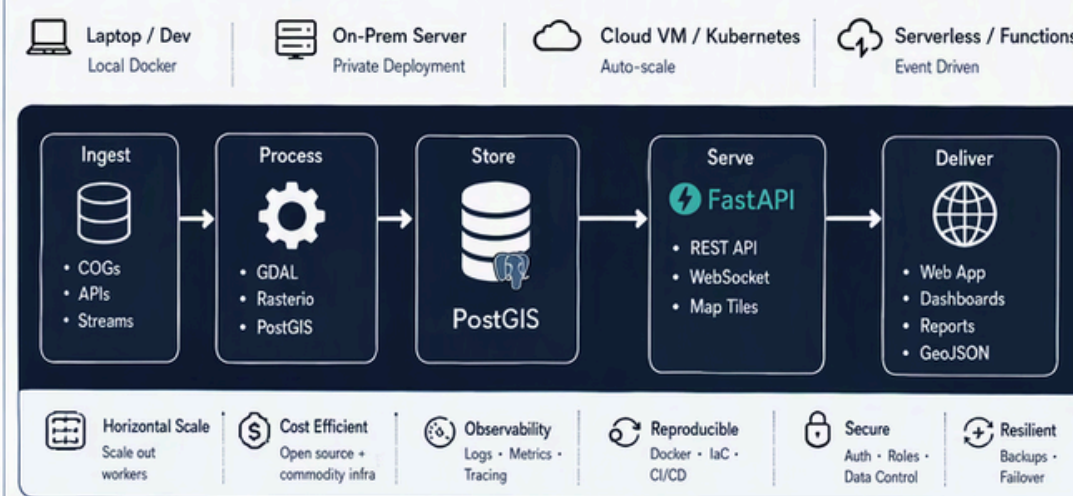
All datasets are free, open and programmatically accessible.

Copernicus
Europe's space for EarthUSGS
science for a changing world

bhoonidhi

4 SCALABLE & DEPLOYABLE ANYWHERE

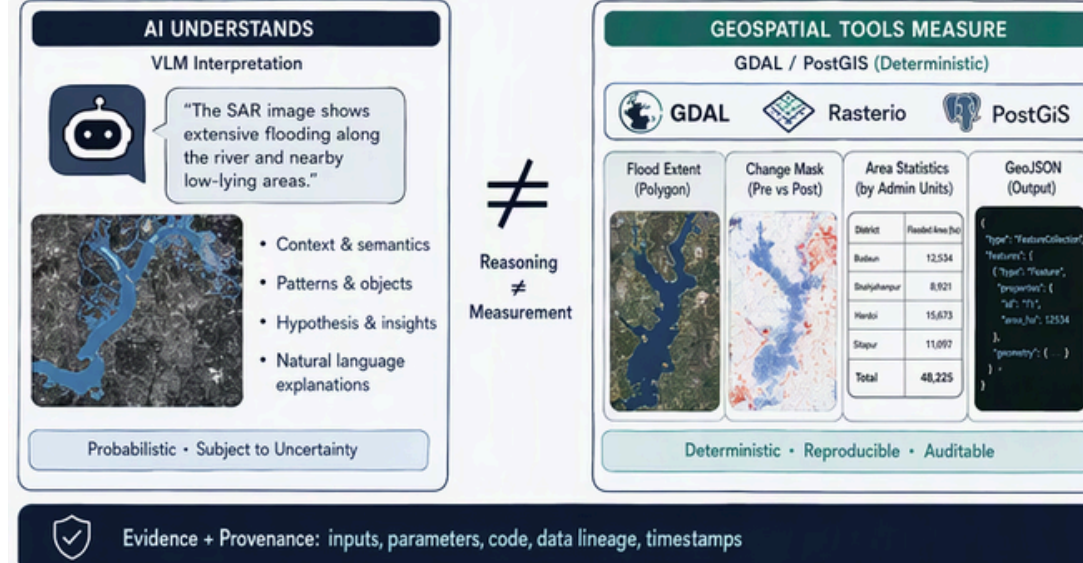
From Laptop to Cloud • Built for Scale



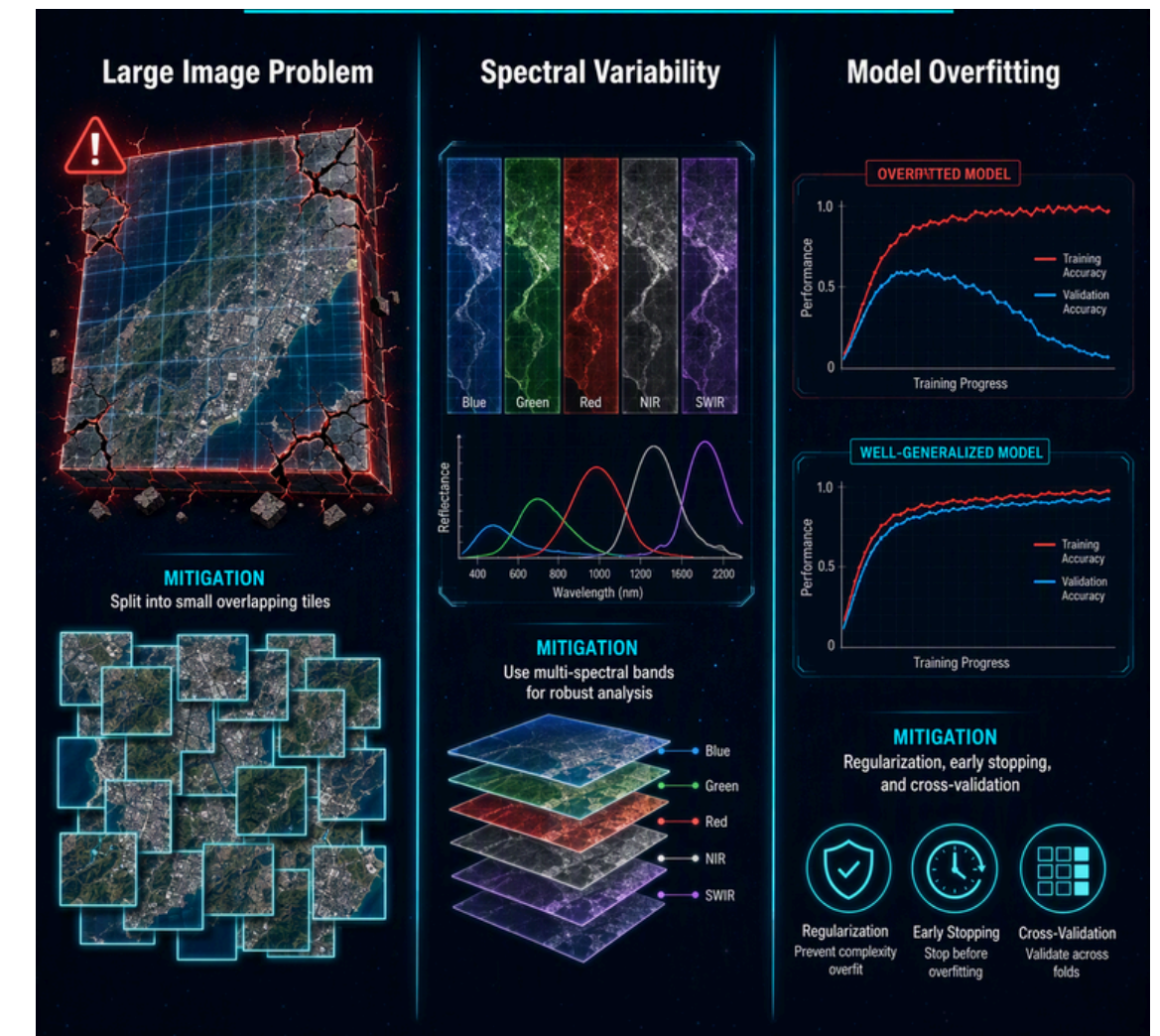
Modular today. Limitless tomorrow.

3 AI + DETERMINISTIC MEASUREMENT

Reasoning ≠ Measurement • Evidence + Provenance

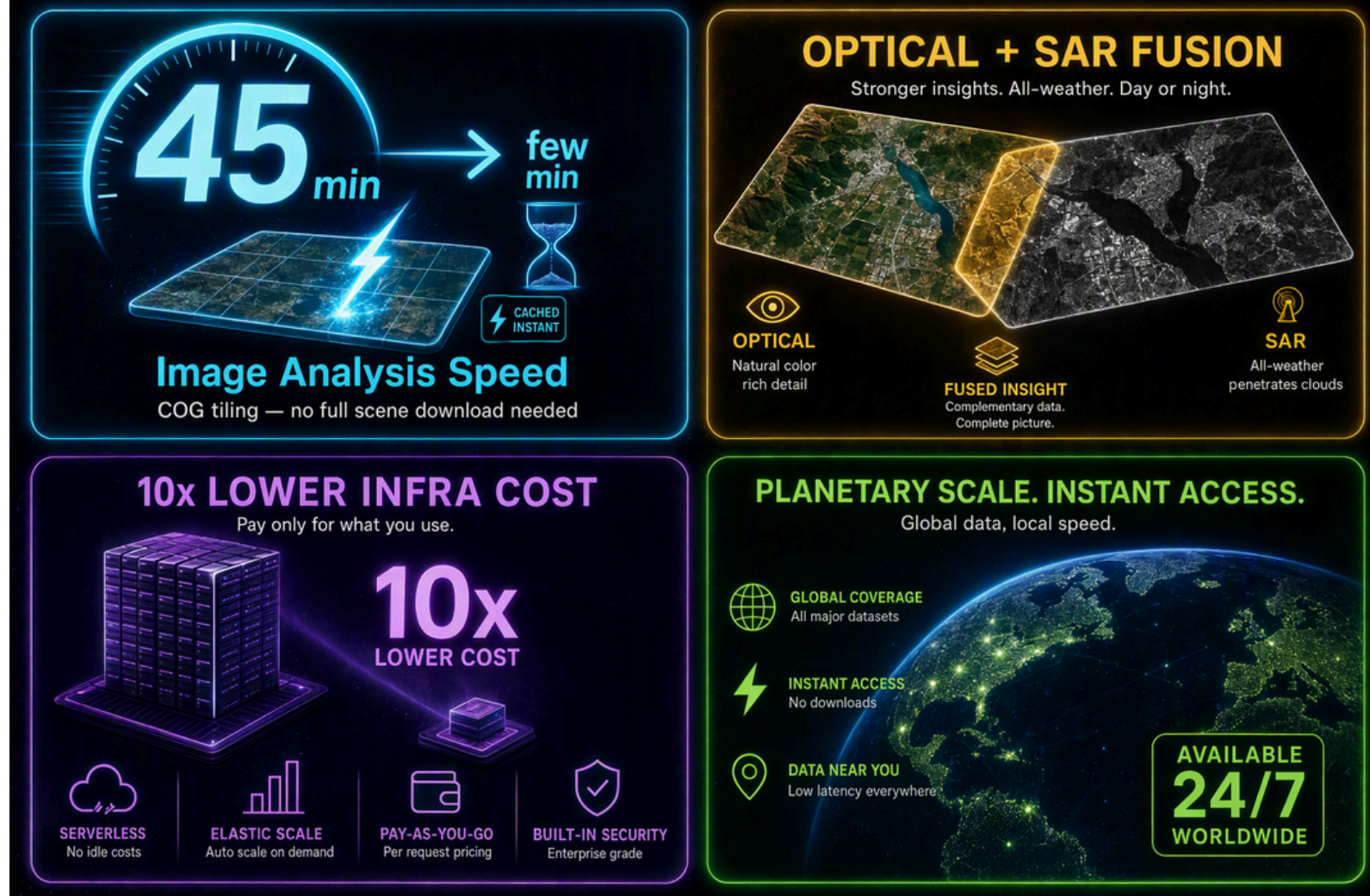


Built with open-source AI, free EO data, and proven geospatial tools, SatQuery AI separates AI reasoning from deterministic measurement to deliver reproducible, auditable results. Its modular architecture supports deployment from a laptop to cloud-scale infrastructure without changing the core foundation.



SatQuery AI addresses three key technical risks in remote-sensing AI. Large satellite scenes are divided into small overlapping tiles to preserve spatial detail while keeping inference efficient. Differences in spectral characteristics across sensors, bands, and acquisition conditions are handled through multispectral/multimodal inputs and robust preprocessing. To reduce model overfitting, we use domain adaptation, regularization, early stopping, and cross-validation across diverse remote-sensing datasets. Together, these measures improve scalability, generalization, and reliability of the system across real-world satellite imagery.

Potential Impact



SatQuery AI can turn a time-consuming GIS workflow into a few-minute query, using COG-based tiling and cached imagery instead of downloading entire satellite scenes. By combining Optical + SAR data from the beginning, it can identify urban and agricultural changes with reported 78–92% confidence. Its core workflow relies on open satellite data and datasets, keeping data costs low, while the natural-language interface lets non-GIS users ask questions directly and understand how the system reached its result through an execution trace.

Benefits of the Solution

- **Easy to use:** Users can interact with the system using plain language without having to learn GIS tools.
- **Better task-specific performance:** Specialist models are designed for specific satellite image tasks and perform better than a general-purpose VLM across benchmarks such as RSVQA, VRSBench, and CDVQA.
- **Less manual work:** The system reduces the time spent manually scanning, analysing, and tagging satellite imagery.
- **Faster responses:** Queries can be processed quickly, which is especially useful for time-sensitive situations such as disaster response.

Easy to Use | Task-Specific AI | Faster Analysis | Evidence-Backed Results | Transparent & Auditable



The growing EO market creates a strong need for simple, practical ways to use satellite data. SatQuery AI brings this capability to areas like agriculture, disaster response, urban planning, and infrastructure, helping government teams, businesses, and researchers get useful insights from satellite images simply by asking questions in natural language, without needing advanced GIS skills.

Data Sources (Live & Training)

Data Sources

- ISRO Bhoonidhi: Sentinel-1/2, Landsat, ResourceSat-2, EOS-04
- Free <5 m GSD baseline data
- NSIL (Demo only): Cartosat-2S/3 and CartoDEM
- Commercial high-resolution scenes

Training Datasets

- BigEarthNet: 590K Sentinel-1/2 pairs → Optical–SAR fusion
- RSVQA: Satellite image question answering
- VRSBench: Region/object grounding
- CDVQA: Temporal change question answering

Research References

Vision–Language Models

- **RS5M & GeoRSCLIP** : Large-scale remote-sensing image–text dataset and domain-adapted VLM [Paper](#) | [Code & Models](#)
- **GeoPixel**: High-resolution RS multimodal model with pixel-level grounding [Paper](#) | [Models](#)

Change Detection

- **ChangeFormer**: Transformer-based bi-temporal change detection [Paper](#) | [Code](#)
- **BIT (Bitemporal Image Transformer)** [Paper](#) | [Code](#)

Evaluation Benchmarks

- **RSVQA** : Remote-sensing Visual Question Answering benchmark : [Paper](#)
- **VRSBench**: [Paper](#) | [Official Project Page](#)

Earth Observation Data

- **Bhoonidhi** : [Official Portal](#)
- **Bhoonidhi Data Policy / Access** : [Bhoonidhi Data Access Policy](#)

